

■ RUBE GOLDBERG MACHINE

Detailed Lesson Plan & Teacher Guide

Subject	STEM / Science & Engineering
Grade Level	Classes 3 – 5 (Ages 8–11)
Duration	60–75 minutes (can be split into 2 sessions)
Group Size	Pairs or groups of 3–4 students
Main Goal	Design and build a 3-step Rube Goldberg machine that performs a simple task using a chain reaction
Curriculum Link	Cause & Effect, Forces & Motion, Sequential Reasoning, Engineering Design Process

■ LEARNING OBJECTIVES

1	Understand Chain Reactions Students will explain that a chain reaction occurs when one event triggers another in sequence.
2	Apply Cause & Effect Students will identify the cause and effect relationship between each step of their machine.
3	Sequence Events Students will use First → Then → Next → Finally language to describe their machine's steps.
4	Engineer a Solution Students will plan, build, test, and improve a simple Rube Goldberg machine using everyday materials.
5	Embrace Failure as Learning Students will understand that failed attempts are part of the engineering design cycle.
6	Expand Vocabulary Students will define and correctly use: chain reaction, cause, effect, sequence, Rube Goldberg.

■ MATERIALS NEEDED

Per Group	Classroom Shared
• 2–3 hardcover books (for ramps) • 1 small ball or marble • 3–4 blocks OR steel tumblers • 1 small ghanti / toy car / flag (finale object) • Ruler or pencils (for levers/barriers) • Masking tape	• Extra cardboard pieces • Parle-G packets (domino stand-ins!) • Empty plastic water bottles • Yarn / string • Sticky notes for labelling steps • Scissors & spare tape rolls

■ LESSON STRUCTURE

PHASE 1 Hook & Wonder Question 10 min

- Display Slide 1 (title). Ask: "Has anything ever fallen and knocked something else over by accident?" Let 2–3 students share.
- Show Slide 2 – WONDER QUESTION: "Can you ring a bell WITHOUT using your hands?"
- Give students 60 seconds of silent thinking. Then take ideas (string, a ball, wind, etc.).
- Tell students: "Today you will actually BUILD something that rings a bell for you!"

PHASE 2 Concept Building – Chain Reaction 12 min

- Show Slide 3. Demonstrate a live domino-style fall using 4–5 books or blocks.
- Ask: "What happened when I pushed the first one?" Build the vocabulary: cause → effect.
- Show Slide 4 (Cause and Effect). Walk through each example. Have students repeat: "When I push THIS... THAT happens!"
- Use Slide 5 – Arjun's Biscuit Story. Read aloud dramatically. Have students count the steps on fingers.
- Check comprehension: "How many things happened because of ONE action?" (Answer: four)

PHASE 3 Sequencing – First → Then → Next → Finally 8 min

- Show Slide 6. Introduce the four sequencing words with actions: clap for FIRST, snap for THEN, stomp for NEXT, jump for FINALLY.
- Do the marble example together using actions as you say each word.
- Quick pair activity: partner A mimes each step while partner B says the word.
- Explain: "Your machine MUST have at least 3 steps in this order."

**PHAS
E 4**

What is a Rube Goldberg Machine?

5 min

- Show Slide 7. One-sentence definition: a chain reaction machine that does a simple job in a long, funny way.
- Show Slide 8 – Simple Tasks. Have students vote on which finale task their group wants.
- Set the challenge: minimum 3 steps, must start with a ball/marble roll, must end with the chosen finale.

**PHAS
E 5**

Build Time!

20 min

- Show Slide 9 (Let's Build!). Distribute materials by group.
- Step 1 (5 min): Groups sketch their design on a sticky note – label each step with First/Then/Next/Finally.
- Step 2 (12 min): Build! Teacher circulates, asking guided questions: "What is the CAUSE here? What is the EFFECT?"
- Step 3 (3 min): Practice run before official test – adjust spacing and ramp angles.
- TEACHER TIP: Resist fixing it for them! Ask: "What do you think went wrong? What would you change?"

**PHAS
E 6**

Test, Reflect & Share

10 min

- Show Slide 10 (Test Your Machine!). Countdown 3-2-1 as a class for each group.
- If it works: celebrate! Ask the group to narrate it using First/Then/Next/Finally.
- If it doesn't: celebrate LOUDER! Say: "You are now REAL engineers – let's debug it!" Give 2 minutes to fix one thing.
- Quick share: each group holds up their machine and names the finale task.

**PHAS
E 7**

Vocabulary & Closing

5 min

- Show Slide 11 (Vocabulary Time). Class calls out definitions while teacher points to each word.
- Show Slide 12 (You Are An Inventor!). Mention real-world connection: ISRO engineers test and retry rockets.
- Assign Slide 13 homework: find one chain reaction at home and draw it in notebook.

■ VOCABULARY REFERENCE CARD

Chain Reaction	A series of events where each one causes the next to happen.
Cause	The action that makes something happen.

Effect	What happens as a result of the cause.
Sequence	The order in which events happen: First → Then → Next → Finally.
Rube Goldberg Machine	An intentionally complex contraption that performs a simple task through a chain reaction.
Engineer	A person who designs, builds, tests, and improves things to solve problems.

■ DIFFERENTIATION STRATEGIES

Support (Struggling Learners)	On Grade Level	Extension (Advanced)
Pre-built ramp provided. Task card with picture-based steps. Partner reads aloud. Limit to 3-step machine.	Build with provided materials. Use sequencing words to narrate. Identify cause/effect for each step.	Add 5+ steps. Add a challenge element (string or pulley). Write a story about their machine character. Record a short video narration.

■ ASSESSMENT & SUCCESS CRITERIA

Chain Reaction	Cannot explain the concept	Explains with prompting	Explains clearly & independently
Cause & Effect	Cannot identify pairs	Identifies with support	Correctly identifies all steps
Sequencing Language	Does not use sequence words	Uses 1–2 sequence words	Uses all 4 words correctly
Machine Design	Machine has fewer than 3 steps	3 steps, partial success	3+ steps, finale achieved
Engineering Mindset	Gives up after failure	Tries one fix with prompting	Self-debuggs and improves

■ HOMEWORK CHALLENGE

Tonight, students become **Chain Reaction Detectives!**

Task: Find ONE chain reaction that happens at home (accidentally or on purpose). Draw it in your notebook using the sequence framework:

FIRST → (what started it?)

THEN → (what happened next?)

NEXT → (what followed?)

FINALLY → (what was the end result?)

Examples students might observe: a cup falls and water spills onto a phone; a bag falls and knocks over a bottle; a door closes and wind blows a paper off a table.

Bring the drawing to share in the next class. Groups that share earn a 'Chain Reaction Star' sticker!

■ TEACHER TIPS & TROUBLESHOOTING

Spacing is everything	The most common reason machines fail is that objects are placed too far apart. Coach students to test marble roll distance before committing to placement.
Noise management	Build time can get loud! Set a noise signal (e.g., ring a small bell) and have students freeze. Ironically appropriate for this lesson.
Safety first	Marbles can roll away fast. Designate build zones on desks with masking tape borders. Remind students not to roll marbles off table edges.
Handling frustration	If a group is stuck, ask: "What is the ONE thing you want to change?" One variable at a time prevents chaos-rebuilds.
Celebrate process, not just success	Take a photo of each group's machine at a key moment. Print and display: "Our Engineering Journey."
Cross-curricular extension	Link to Hindi/language arts: have students write a 5-sentence story narrating their machine's "adventure." Link to math: measure the distance each ball rolls using a ruler.